

Math 421

Friday, September 19

Announcements

- Homework 2 due today – make sure the submission has gone through and appears in gradescope!
- Information about makeup midterms have been posted – requires prior approval.

Proof Practice

Theorem. If x is a rational number and y is an irrational number, then $x + y$ is irrational.

Discussion.

- Determine proof method.
Contradiction
- What are the hypotheses
- Use definitions.
 - $x \in \mathbb{Q}$
 - $y \notin \mathbb{Q}$
 - $x + y \in \mathbb{Q}$
 - $x = \frac{p}{q}, q \neq 0$
 - $x + y = \frac{r}{s}, s \neq 0$
- Sketch out necessary algebra.

$$x + y = \frac{r}{s} \Rightarrow \frac{p}{q} + y = \frac{r}{s} \Rightarrow y = \frac{r}{s} - \frac{p}{q}$$

Proof. We will use a proof by contradiction. Let $x \in \mathbb{Q}$, $y \notin \mathbb{Q}$, and $x + y \in \mathbb{Q}$. Thus $x = \frac{p}{q}$, $x + y = \frac{r}{s}$ for $p, q, r, s \in \mathbb{Z}$, $q, s \neq 0$. Therefore,
$$x + y = \frac{p}{q} + y = \frac{r}{s} \Rightarrow y = \frac{r}{s} - \frac{p}{q}$$
$$\Rightarrow y = \frac{rq - ps}{sq}. \quad rq - ps \in \mathbb{Z},$$
$$sq \in \mathbb{Z}, \quad sq \neq 0. \quad y \in \mathbb{Q}.$$

Contradiction.

Group Activity – Proof Practice

Theorem. If $x, y \in \mathbb{R}$, then $|xy| = |x||y|$.

Form groups of 4-6. Further divide in halves **A** & **B**. Each half proves 2 of the cases. Then regroup and compare.

Proof.

Case 1: Assume $x \geq 0$ and $y \geq 0$. Then $xy \geq 0$. So $|xy| = xy = |x||y|$

Case 2: Assume $x < 0$ and $y < 0$. Then $xy > 0$. So $|xy| = xy = (-x)(-y)$
 $|x| = -x$ $|y| = -y$
 $= |x||y|$.

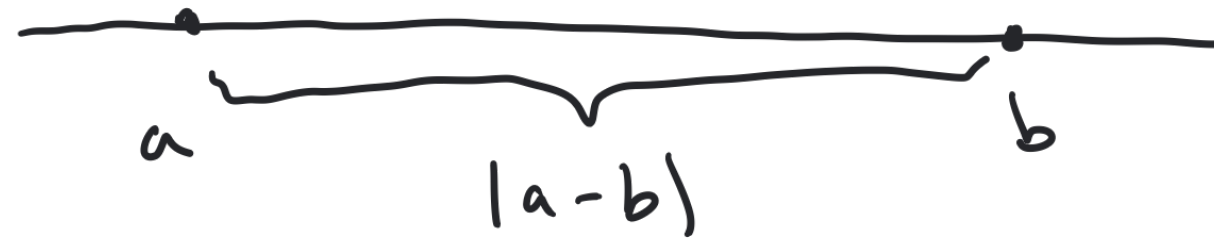
Case 3: Assume $x \geq 0$ and $y < 0$. Then $xy \leq 0$. So $|xy| = -(xy)$
 $= x(-y) = |x||y|$.

Case 4: Assume $x < 0$ and $y \geq 0$.

By swapping variables, this follows from case 3.

Distances

If $a, b \in \mathbb{R}$, then $|a - b|$ is the distance between a and b .



Notice that if $c \in \mathbb{R}$, then

$$|a - b| = |(a - c) + (c - b)| \leq |a - c| + |c - b|$$

adding 0

Proof Practice

Thm. Suppose $a, b \in \mathbb{R}$. Then $a = b$ if and only if for every positive real number $\varepsilon > 0$ we have $|a - b| < \varepsilon$.

Discussion.

- ❖ What is this theorem saying in your own words? Do you believe it?
- ❖ What are the two conditional statements we will have to prove?

$$\begin{aligned} (\Rightarrow) \quad & \text{If } a = b \text{ then } \forall \varepsilon > 0, |a - b| < \varepsilon. \\ & a = b, |a - b| = 0 < \varepsilon \quad \forall \varepsilon > 0. \end{aligned}$$

Proof Practice $\forall \varepsilon > 0$

Proof. ($\Leftarrow$) Suppose $|a-b| < \varepsilon$. Assume, by way

of contradiction, $a \neq b$, $|a-b| = \varepsilon_0 > 0$.

By assumption $\varepsilon_0 = |a-b| < \varepsilon_0$, $\varepsilon_0 < \varepsilon_0$ is

a contradiction.